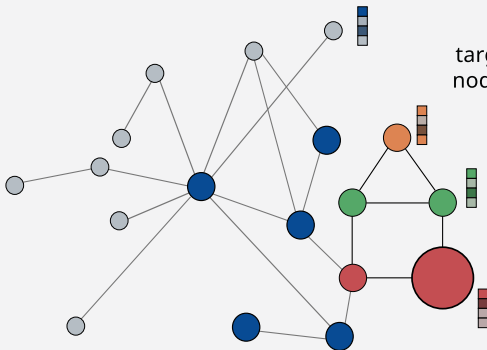


Clexplainer:

L-hop neighbourhood of target node

feature vector
target explanation node

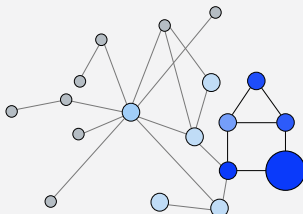


Explanation Subgraph:

Take maximum Causal effect for each node:

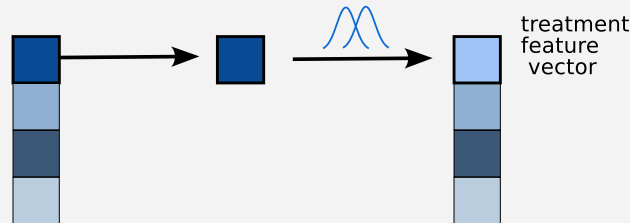


Choose top-k causal effect nodes as part of explanation subgraph:



Counterfactual generation:

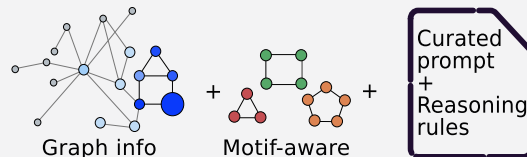
For each feature in the node:



Calculate causal effect:

$$\tau_{w,i}(a) = Y_{t(a)}(\mathcal{G}, v, w, i) - Y_c(\mathcal{G}, v, w, i).$$

Textual Explanation:



=

Textual explanation of the explanation subgraph for a given node. Only positive nodes, or in the multi-class case (1,2,3) have an explanation sub-graph. Nodes in blue do not have an explanation sub-graph. A similar procedure occurs for graph classification and link prediction, such that in the graph classification case there is one explanation sub-graph per graph. And for link prediction, the explanation sub-graph exist for existing links.